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13

ONE-DIMENSIONAL UNSTEADY FLOWS

13.1 General program for unsteady flows

Program DSMC1U.FOR is a version of DSMC1 that has been adapted to make ensemble averages of repeated runs of an unsteady flow, rather than a time average of a steady flow that is established as the large-time state of a single run of an unsteady flow. As in the preceding programs, the flow is sampled each NIS time steps, but a separate sampling record must be made for each of these sampling events. Excessive memory would be required if the sampling arrays were simply given an extra dimension equal to the total number of sampling events MNOI in each run. The answer is to establish the direct access file DSMC1U.DAT with MNOI records. Note that MNOI is set in the PARAMETER statement and, together with NIS, it controls the length of each run. No more than one record of this file is in memory at any time and, on some occasions that a record of this file is read, most of the variables must be read into dummy arrays so that they do not overwrite the current values of these variables that already exist.

The 'restart' file is not needed because a restart of the program simply adds to the number of runs, and each run starts from the initial conditions specified in the subroutine DATA1U. The output file DSMC1U.OUT now includes the results at each of the sampling times. This amounts to a great deal of information and, if this is to serve as the primary output, the number of sampling times will be restricted to as small a number as possible. On the other hand, if a post-processing program is to be used to automatically construct contours in the distance–time plane, it is desirable to have output at a large number of times. The results in this chapter were obtained through a post-processing program and further details are given in Appendix F.

In order to cope with the desired range of problems, several additions have been made to the range of available boundary conditions. The first is that the uniform initial gas need not extend over the whole of the flowfield, and there may be an initial interface with a vacuum at some location within the flow. The second and more complex new option is that either boundary may be a moving piston. The piston may be plane, cylindrical, or spherical and the molecules are reflected specularly from its face. The number of cells remains constant and, when one or more piston boundary conditions are chosen, the cell boundaries and volumes become a function of time and must be recalculated at each time step.

In the case of a plane flow, the extensions to cope with the moving piston are quite straightforward. The fraction of the trajectory element to the intersection with the piston is

$$s_d = (X_i - x_i)/\{(u - U)\Delta t\}, \tag{13.1}$$

where $_i$ denotes the value at the beginning of the time step Δt, x and u are the location and x-velocity component of the molecule, and X and U are the corresponding quantities for the piston. The post-reflection velocity component of the molecule is

$$u' = 2U - u. \tag{13.2}$$

For cylindrical and spherical flows, the extension of eqn (12.12) to include the finite piston velocity gives

$$\tfrac{1}{2}s_d^2\{1-(U\Delta t/d)^2\} + (u\Delta t\, x_i - U\Delta t\, r_i)s_d/d^2 + \tfrac{1}{2}(x_i^2 - r_i^2)/d^2 = 0. \tag{13.3}$$

This equation is evaluated in the subroutine RBCP and, after the collision point velocities have been transformed by subroutine AIFX, eqn (13.2) applies also to these cases. Because there can be multiple reflections with a curved piston during a single time step, the reflected motion is calculated as a separate trajectory element.

13.2 The formation of a strong shock wave

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The data in the listing of program DSMC1U is for the formation of a strong plane shock wave by a piston that, at zero time, is impulsively given a constant velocity of 2285.5 ms^{-1} in the x-direction. The gas in front of the piston is the VHS model of argon at a temperature of 273 K and a number density of 10^{20} m^{-3}. The Rankine–Hugoniot theory shows that this leads to a shock of Mach number 10, i.e. $(Ma)_S = 10$. The rate of propagation of this wave is 3078.1 ms^{-1} and the downstream number density and temperature are 3.8835×10^{20} and 8796.6 K, respectively.

A time average was taken over 700 separate runs and the contours, in the distance–time plane, of velocity, number density, temperature, and parallel temperature are shown in Figs 13.1 to 13.4. While the velocity and number density profiles across the fully formed wave are simply the inverse of one another, the formation contours are qualitatively different. The downstream velocity is established immediately, while the density adjacent to the piston increases gradually to the downstream value. The collisionless value of the density adjacent to the piston is given by eqn (7.43) and Fig. 7.5 shows that, for large piston speed ratios, the initial density adjacent to the piston is twice the undisturbed density. Eqn (7.43) also gives the path of the collisionless contours and Fig. 13.2 shows the collisionless contour for $n = 1.2\times10^{20}$. The continuum shock path has also been added to Fig. 13.2, and it can be seen that, except for very small times, it very nearly coincides with the contour for $n = 2.4\times10^{20}$ which is very close to the mean of the upstream and downstream number densities.

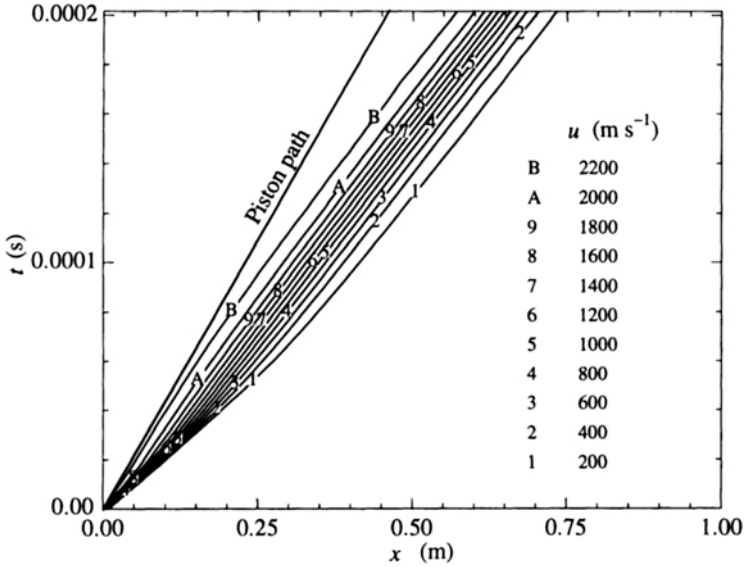


Fig. 13.1 Velocity contours in the distance–time plane.

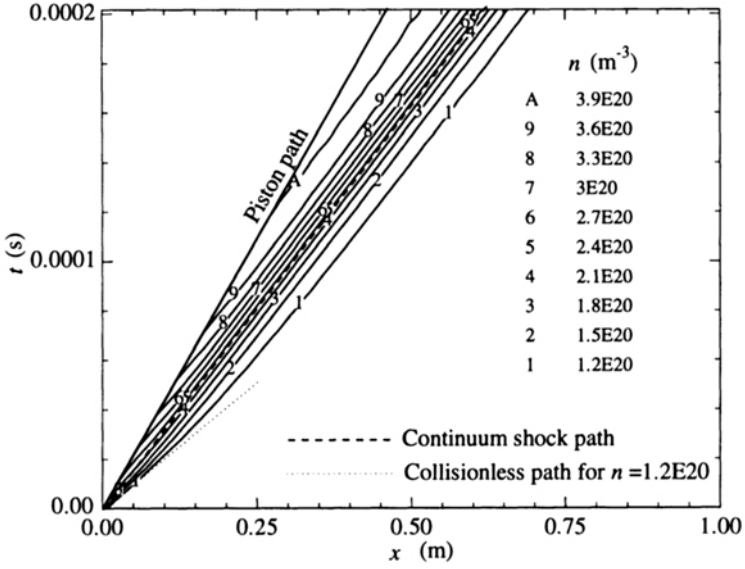


Fig. 13.2 Number density contours in the distance–time plane.

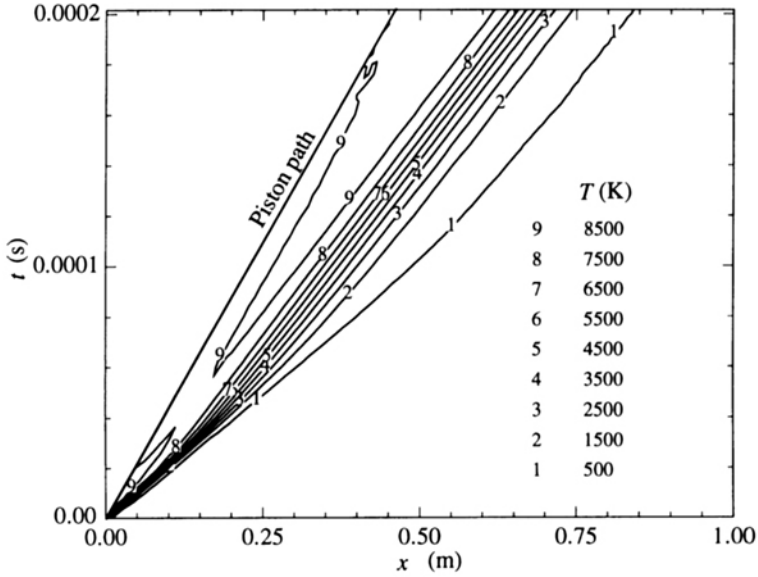


Fig. 13.3 Temperature contours in the distance–time plane.

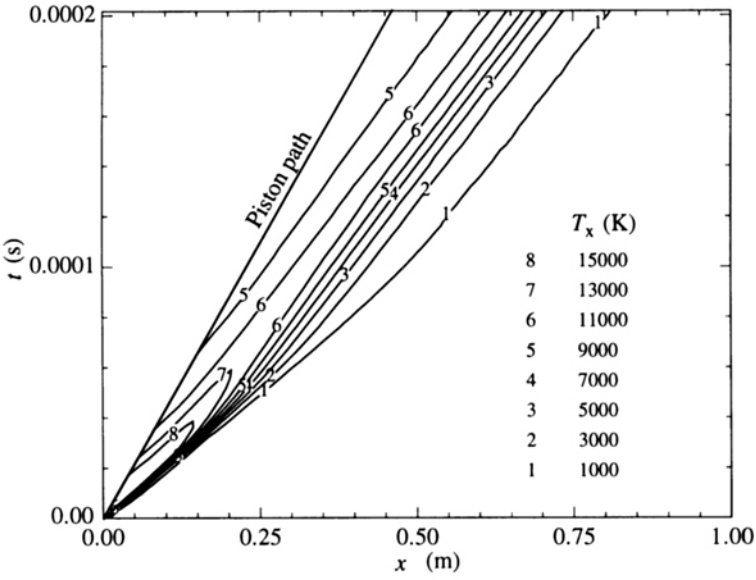


Fig. 13.4 Parallel temperature contours in the distance–time plane.

There is an apparent anomaly in the density contours in that contour A is for a number density of 3.9×10^{20} even though the Rankine–Hugoniot value is only 3.8835×10^{20}. The reason for this is that the partially formed shock involves a greater degree of non-equilibrium than the fully formed shock and the entropy increase across it is greater than that across the fully formed shock. This leads to a lower stagnation temperature in the gas adjacent to the piston than in the gas immediately behind the fully formed shock. This leads in turn to a reduced temperature and therefore an increased density in the gas adjacent to the piston. The 'entropy layer' effect is apparent in the temperature contours of Fig. 13.3. The downstream Rankine–Hugoniot temperature of just under 8,800 K is attained at about 100 μs. The temperature adjacent to the wall gradually increases as thermal conduction reduces the temperature gradient that is associated with the entropy gradient. The extreme non-equilibrium in the early stages of the shock formation is apparent in the parallel temperature contours of Fig. 13.4. The maximum parallel temperature in the fully formed shock is just over 11,000 K , but a large region has a parallel temperature in excess of 15,000 K.

13.3 The reflection of a strong shock wave

There is a stationary specularly reflecting surface at the outer boundary at $x = 1$ and, if the time span of the run is increased, the shock wave reflects from this surface. The continuum Rankine–Hugoniot theory predicts a reflected shock of Mach number $(Ma)_S = 2.2$. The number density and temperature behind the reflected shock are then 9.593×10^{20} and 20,676 K, respectively.

The DSMC results for the shock reflection process are illustrated in Figs 13.5 to 13.8. The minimum and maximum limits in the velocity are unchanged and the contours are continuous. The number density at the wall increases continuously during the reflection process and the density corresponding to that behind the incident shock occurs very close to the predicted point of impingement of the continuum shock. The velocity gradient normal to the wall is a maximum at this time. Although the shock Mach number of the reflected shock is relatively small, the density upstream of the wave is almost four times that of undisturbed gas and the actual thickness of the reflected wave is less than that of the incident wave.

As with the incident shock, the contour of the mean density in the reflected shock is close to the continuum shock path. The temperature profiles of both the incident and reflected waves show the characteristic shock behaviour of the downstream temperature being almost attained upstream of the centre of the density contour. The temperature contours during the reflection process are qualitatively similar to the number density contours. However, the time at which the incident shock contours change to reflected shock contours is earlier than that for the number density profiles.

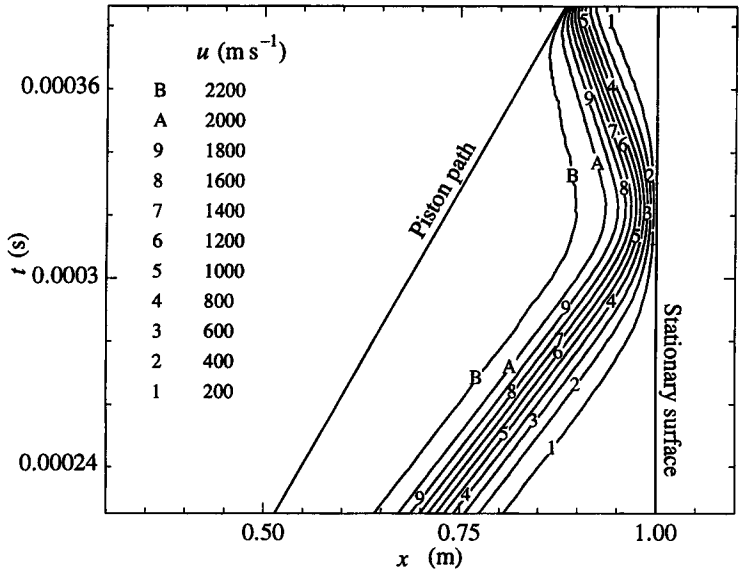


Fig. 13.5 The velocity contours in the shock reflection.

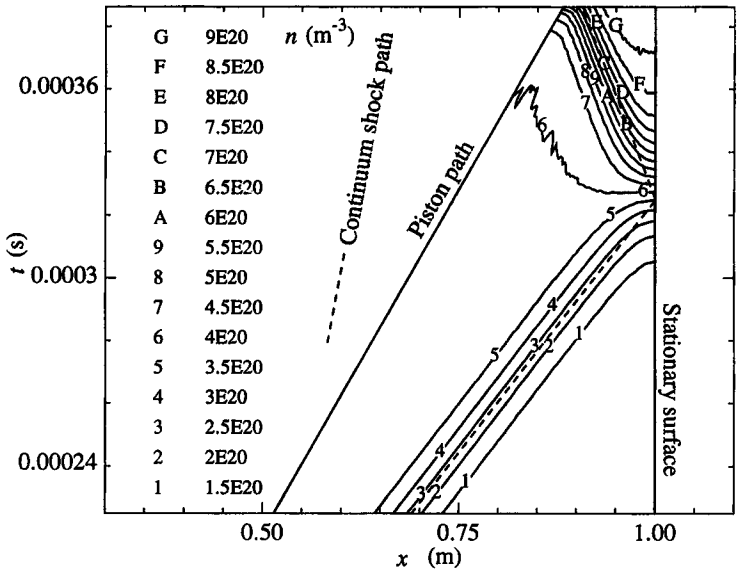


Fig. 13.6 The number density contours in the shock reflection.

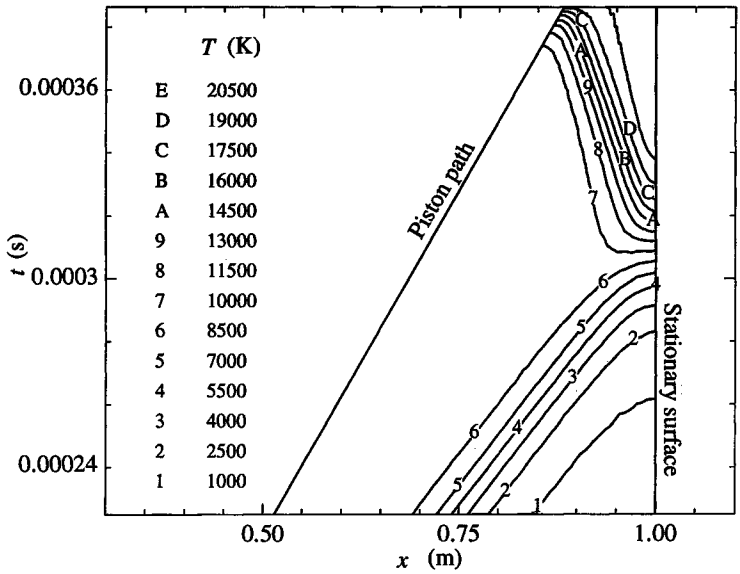


Fig. 13.7 The temperature contours in the shock reflection.

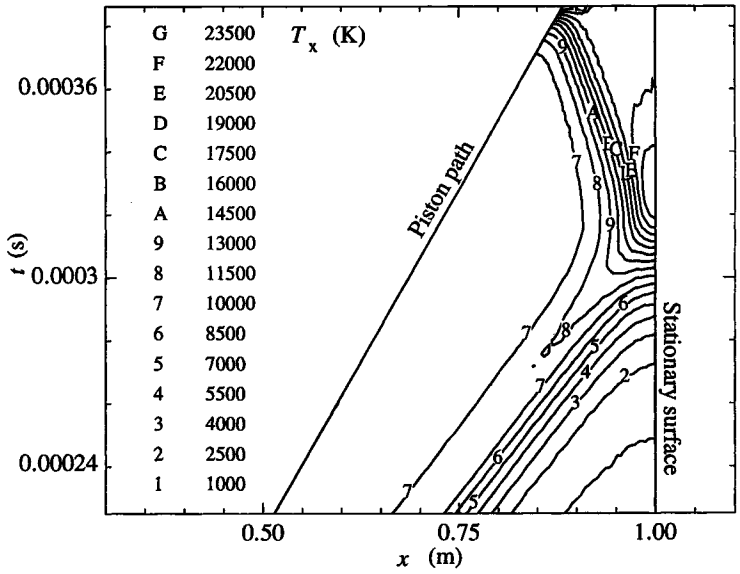


Fig. 13.8 The parallel temperature contours in the shock reflection.

13.4 Formation of a weak shock wave

A calculation similar to that in §13.2 was made for a comparatively weak shock of $(Ma)_S = 1.4$. The required piston speed is 158.3 m/s and the Rankine-Hugoniot values of the number density and temperature are 1.581×10^{20} m^{-3} and 380.0 K, respectively.

The velocity contours are shown in Fig. 13.9. The downstream contour is for a velocity only 2% less than the piston velocity and the statistical scatter is responsible for the jagged appearance of this and, to a lesser extent, the first contour. The 95 m/s contour is almost coincident with the continuum shock path.

The number density contours in Fig. 13.10 are qualitatively different from those for the strong shock and are generally similar to the velocity contours. The reason is that the collisionless density at the piston face from eqn (7.43) is 1.49×10^{20} m^{-3}, and this is only slightly less than the continuum density ratio. This means that the density rise along the piston is quite small, but there is again a density overshoot near the piston as a result of the entropy layer. As in the strong shock case, the maximum value of this overshoot is about 2% . While the Navier-Stokes equations are valid for the structure of a fully formed shock wave of $(Ma)_S = 1.4$, they are not valid for the stronger gradients that occur in the early stages of the shock formation. Finally, the temperature contours of Fig. 13.11 clearly show the effects of the entropy layer and also show that most of the temperature rise occurs ahead of the continuum shock path.

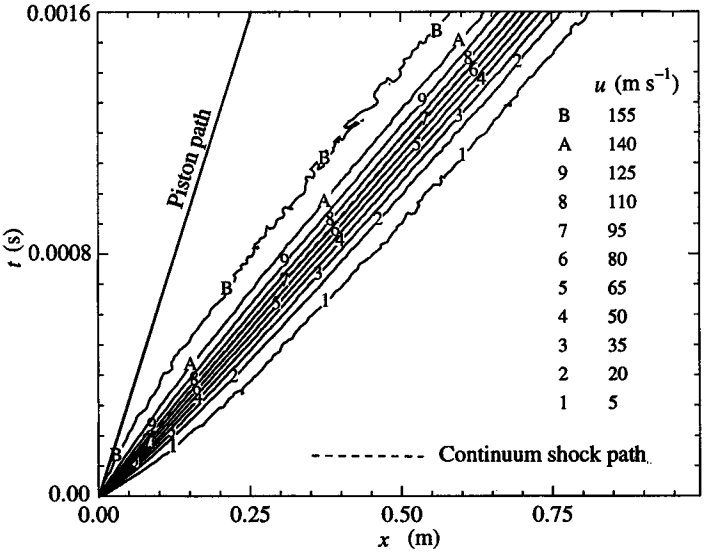


Fig. 13.9 Velocity contours in the formation of a weak shock wave.

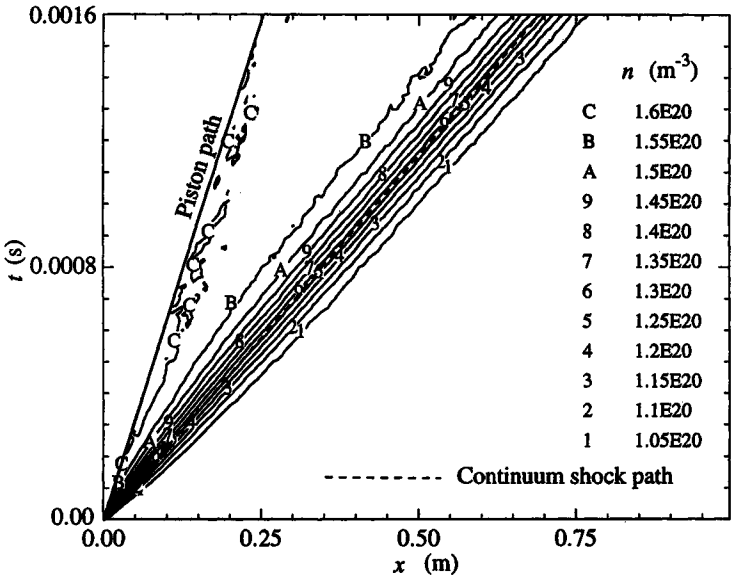


Fig. 13.10 Number density contours in the formation of a weak shock.

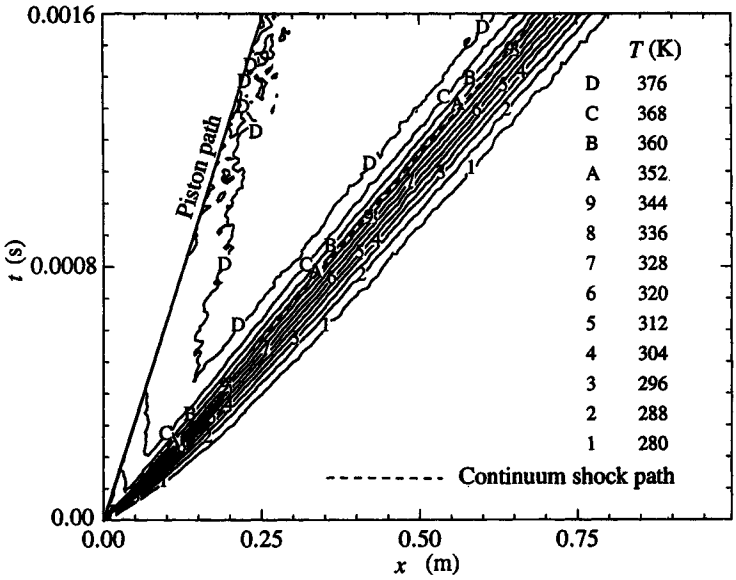


Fig. 13.11 Temperature contours in the formation of a weak shock wave.

13.5 The complete rarefaction wave

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A complete rarefaction wave arises at a discontinuous plane interface between a uniform gas and a vacuum. In the continuum solution, the leading edge of the wave advances into the gas at the speed of sound while, within the wave, the gas expands into the vacuum. The speed of the expanding gas becomes sonic at the location of the interface, and the flow Mach number is infinite at the tail of the expansion even though the limiting speed is only $2/(\gamma-1)$ times the undisturbed speed of sound. The continuum theory must break down at the tail of rarefaction wave at all times and, at very small times, it will fail over the whole flow.

An indication of the expected extent of the breakdown of translational equilibrium in the rarefaction wave may be gained from the breakdown parameter P defined by eqn (1.4). This expression has been combined with the continuum equations for the complete rarefaction (Bird 1970), and the initial value of P at the leading edge of the expansion is

$$P_i = \{1/(\gamma-1)\}(\pi\gamma/2)^{1/2}(\lambda_0/x_i). \tag{13.4}$$

Here, λ_0 is the mean free path in the undisturbed gas and x_i is the distance measured from the initial discontinuous interface. The subsequent change in P along the particle paths is given by

$$DP/Dt = (P/t)\{(\gamma-1)(\gamma+1)\}(1-2\omega). \tag{13.5}$$

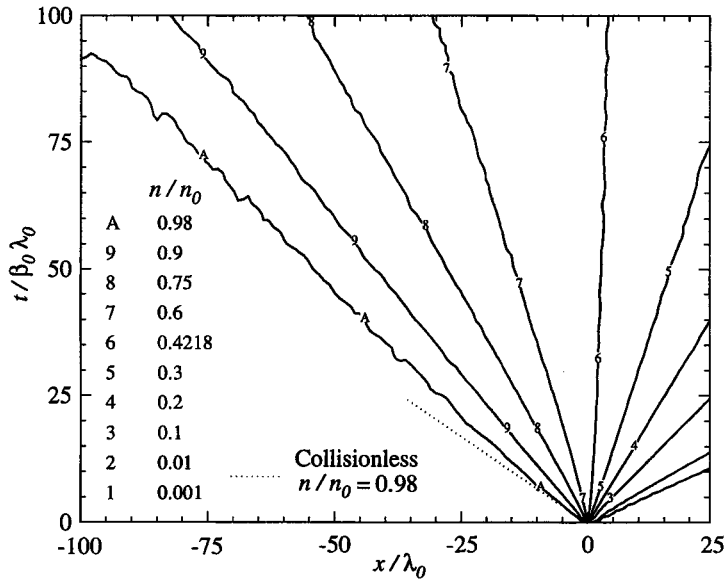


Fig. 13.12 Number density contours in a complete rarefaction wave.

This shows that P remains constant along a particle path in a hard sphere gas but, for more realistic models, there is a gradual reduction in P and a consequent move towards equilibrium. For a monatomic gas, eqn (13.4) reduces to

$$P_i = 0.60676\lambda_0 / x_i. \tag{13.6}$$

Should the values of P that were found in §12.10 to be necessary for a steady expansion apply also to an unsteady flow, the flow at the leading edge of the rarefaction would not be in equilibrium until it had progressed some hundreds of mean free paths into the gas.

Program DSMC1U was used to calculate the expansion of a monatomic gas with an extent of 100 mean free paths from a plane of symmetry. The initial vacuum extended for 50 initial mean free paths and was terminated by a 'vacuum' boundary. Some typical contours of constant number density are shown in Fig. 13.12. These are normalised to the undisturbed number density n_0, and the contour for 0.98 is close to the leading edge of the wave. The initial slope of this contour in the distance-time plane is consistent with the collisionless theory of eqn (7.41). The effective speed of the wave then decreases, but it reflects from the plane of symmetry well before the normalised time of 109.5 that is predicted by the continuum theory. This theory predicts a constant number density ratio of 0.4218 at the origin. This contour has a positive slope, but gradually becomes more vertical.

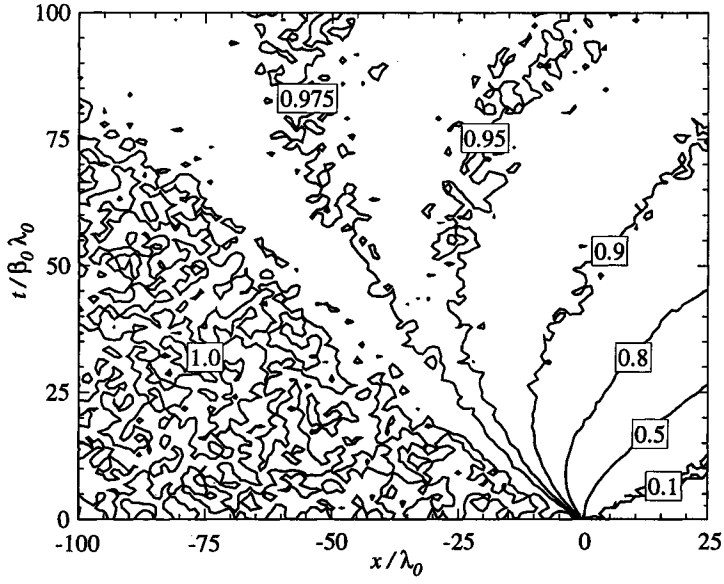


Fig. 13.13 The ratio of the parallel to the normal temperature.

As in the steady expansion, a more precise measure of the degree of nonequilibrium in the expansion is given by the ratio of the parallel temperature T_x to the normal temperature T_n. Contours for this ratio are shown in Fig. 13.13. The ratio is, of course, unity in the undisturbed gas but the statistical scatter produces a random pattern for this contour. Scatter is also present for the values of the ratio near unity, but the figure clearly shows the gradual approach to equilibrium in early part of the expansion together with the extreme nonequilibrium near the tail of the rarefaction wave.

Key contours of velocity normalised to the undisturbed speed of sound a_0 are shown in Fig. 13.14. The contour for 0.02 is consistent with the number density contour of 0.98, except that the reflection of the wave at the plane of symmetry causes an outward rather than inward deflection of the contour. The collisionless theory of eqn (7.42) predicts an initial velocity ratio of 0.618 at $x=0$, but this applies only at extremely small times. The continuum prediction of sonic flow at this location leads in a monatomic gas to a velocity ratio of 0.75. This contour is similar to the density contour for sonic flow. These contours would become vertical at a normalised time of 200 at only five or six mean free paths from $x=0$. Despite the extreme temperature nonequilibrium, the limiting velocity of $3a_0$ is attained at the tail of the rarefaction (or front of the advancing gas).

The results are generally consistent with the theory for the behaviour of the breakdown parameter, with a gas element that is initially out of equilibrium tending to remain out of equilibrium along its particle path.

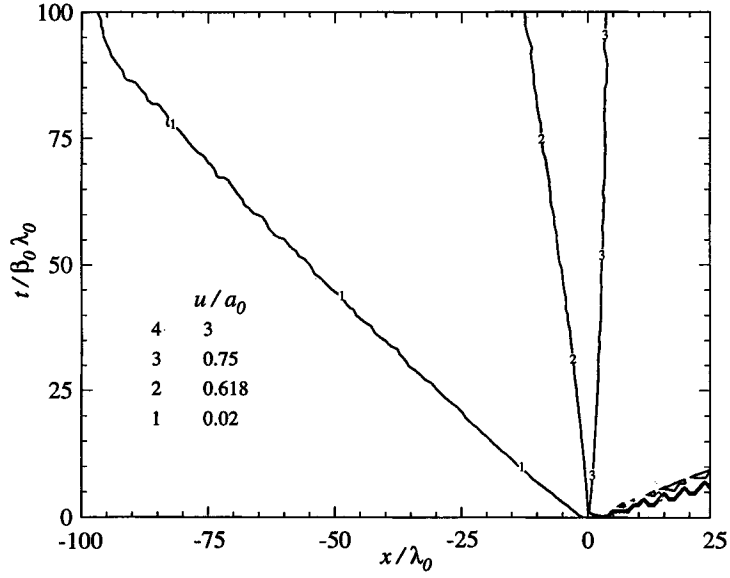


Fig. 13.14 The key velocity contours in the rarefaction wave.

13.5 Spherically imploding shock wave

In the plane flow case, the shock waves are of constant strength and the flow conditions are uniform outside the waves. This ceases to be the case for waves in cylindrically or spherically symmetric flows. The reflection of a spherical wave from the centre or origin has been the subject of many studies because the continuum theory with a discontinuous wave leads to a singularity at the origin. The singularity disappears for real shock waves of finite thickness, and it is necessary to define the problem by the process that produces the wave and the strength of the wave when it is a specified number of mean free paths from the origin. The first case to be considered is a wave in argon that is driven by a hollow spherical piston that is contracting with a uniform velocity of -2285.5 ms^{-1}. The initial gas number density is 10^{20} m^{-3} and the contraction starts impulsively from a radius of 1 m. This corresponds to a shock Mach number of 10 at a radius of approximately 70 mean free paths.

There is a problem in obtaining a good sample near the origin. The calculation was made with 30,000 simulated molecules and the width of the cell at the origin was one hundred times the width of the cell adjacent to the piston. While this led to a reasonably large sample in each cell, the inner tenth of the flow in terms of the radius initially contained only 300 molecules. This caused an unexpected problem that required modifications to the code.

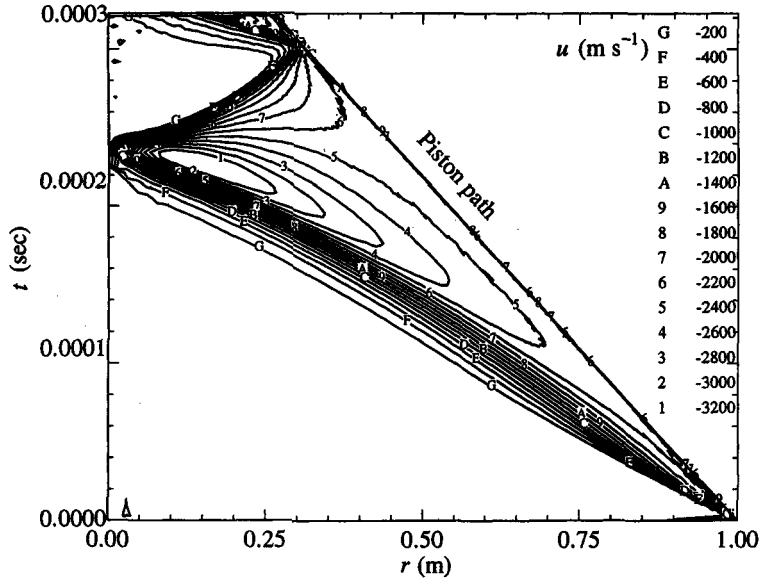


Fig. 13.15 Velocity contours in a spherically imploding shock wave.

In all the preceding programs that involve the generation of an initially uniform gas, each of the molecules in that gas is generated independently. This means that there are always initial non-uniformities in the gas and these lead to extraneous disturbances. These have not been a problem in the earlier applications but, in this case, the disturbances that move towards the axis were amplified to such an extent that the flow near the axis was seriously affected. The disturbances had the characteristics of pressure waves in that both the density and temperature were either increased or decreased, and their timing was such that they came from the inner 300 molecules. The subroutine INIT1U in program DSMC1U was modified such that the initial molecules were generated in pairs rather than as single molecules. The two molecules in each pair had radial velocity components that were equal and opposite, but were otherwise identical. There was therefore no net initial momentum in the radial direction and the problem disappeared. This modification is in the nature of a 'quick fix' and there is a need for procedures that systematically generate initial states that are, as near as possible, unbiased in all respects. This is one aspect of the 'variance reduction' issue that has been discussed in §7.7 and in §10.4. Given the large number of dimensions of phase space, this will probably be best achieved through matched lists rather than through uniform spacing of the molecules in phase space.

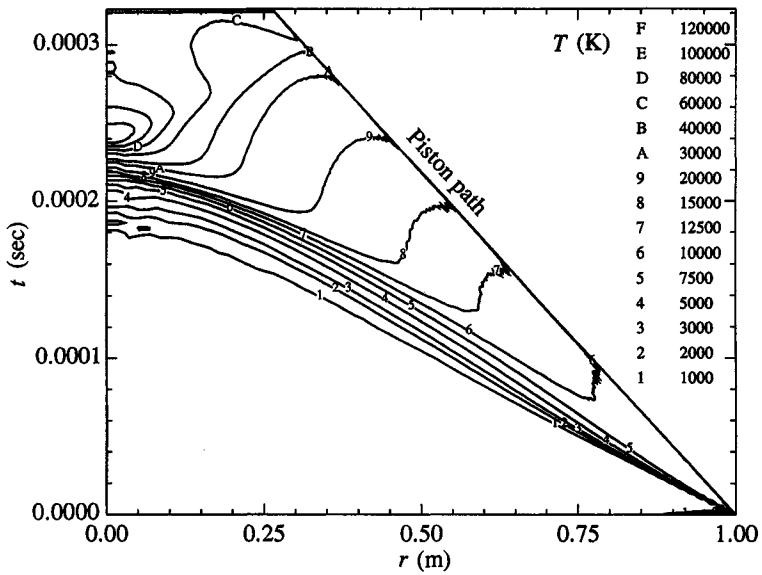


Fig. 13.16 Temperature contours in a spherically imploding shock wave.

The velocity contours of Fig. 13.15 show that, as expected, the speed of the incident shock increases as it moves toward the origin and that of the reflected shock decreases as it moves away from the origin. The velocity of the gas behind the incident shock and ahead of the reflected shock is higher than the piston velocity by a maximum of about 50%. On the other hand, the gas behind the reflected shock is almost stationary.

The magnitude of the maximum temperature at the origin has been the focus of much of the interest in this flow. Figure 13.16 shows that it is just over 150,000 K in this case, but ionization has not been included in the simulation and this would significantly reduce the maximum temperature in the real gas. Also, the simulation procedures force the simulated flow to be spherically symmetric while asymmetries could develop in a real flow. The most interesting feature of the temperature contours is that the maximum temperature occurs after, rather than during, the formation of the reflected shock.

The approximate theory of Whitham (1958) predicts that the shock Mach number of a strong spherical shock wave is inversely proportional to the radius raised to the power $\mu\gamma/\{2+\mu)(\mu+\gamma)\}$, where $\mu^2 = 2\gamma/(\gamma-1)$. The resulting shock path is included in Fig. 13.17 for the density contours in the distance–time plane. The theory is in reasonable agreement with the DSMC calculation. Note that the maximum density at the centre coincides with the temperature maximum.

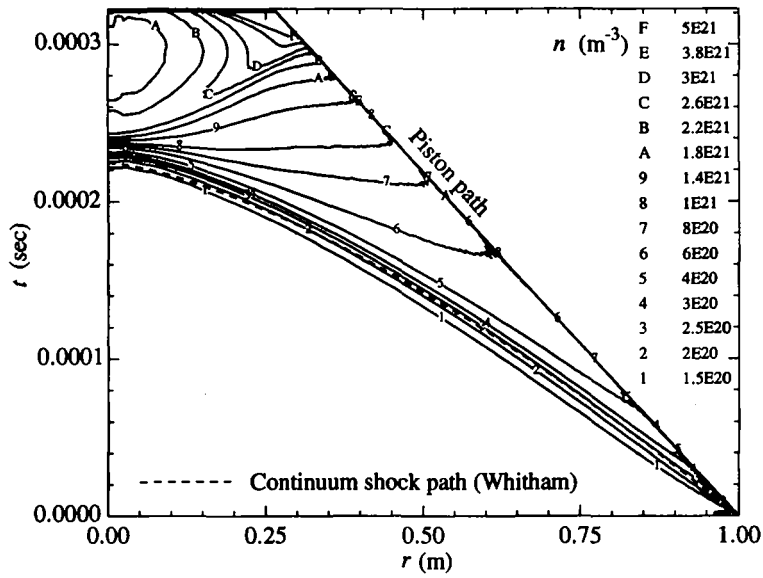


Fig. 13.17 Density contours for the spherically imploding shock wave.

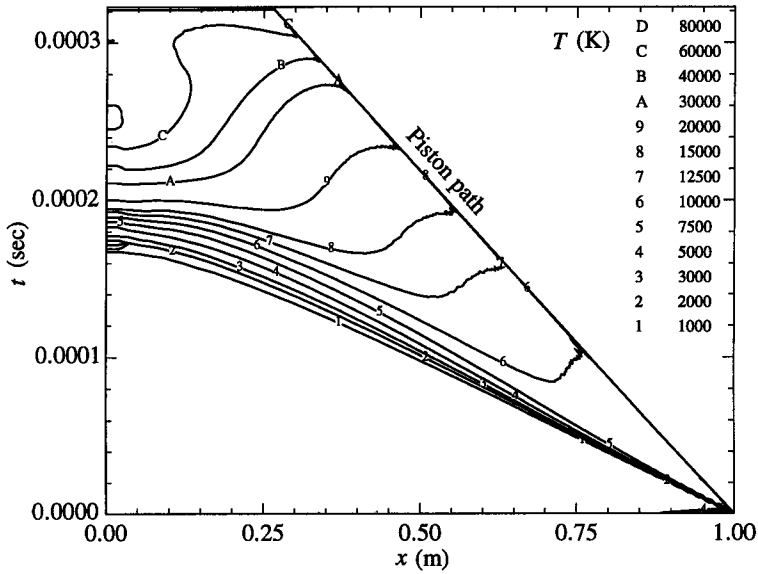


Fig. 13.18 Temperature contours at a density half that of Fig. 13.13.

If the number of mean free paths between a shock of given strength and the origin is increased or decreased, the temperature maximum in the reflection at the origin would be expected to also increase or decrease. In order to investigate this effect, the calculation was repeated for an initial number density of 0.5×10^{20}. The temperature contours in the distance–time plane are shown in Fig. 13.18. The reduction in density by a factor of two causes the maximum temperature to fall by one third.

An interesting point is that the reduction in density causes the shock profile to be sharper at small times. This is because the profile becomes wider with time during shock formation and the reduction in the collision rate slows this process.

13.6 Collapse of a cylindrical cavity

The final example is the flow that is produced when, at zero time, a cylindrical cavity with a radius of 0.5 m exists in argon of number density 10^{20} m^{-3} at temperature 273 K. A full rarefaction wave is initially centred at the boundary between the gas and the vacuum. The edge of this moves at the undisturbed speed of sound, but the cylindrical geometry causes the other characteristics in the distance–time plane to be curved. The high-speed tail of the rarefaction travels towards the axis of the flow and eventually reflects from it. This leads to the compression of the gas near the centre by compression waves that eventually move out from the axis.

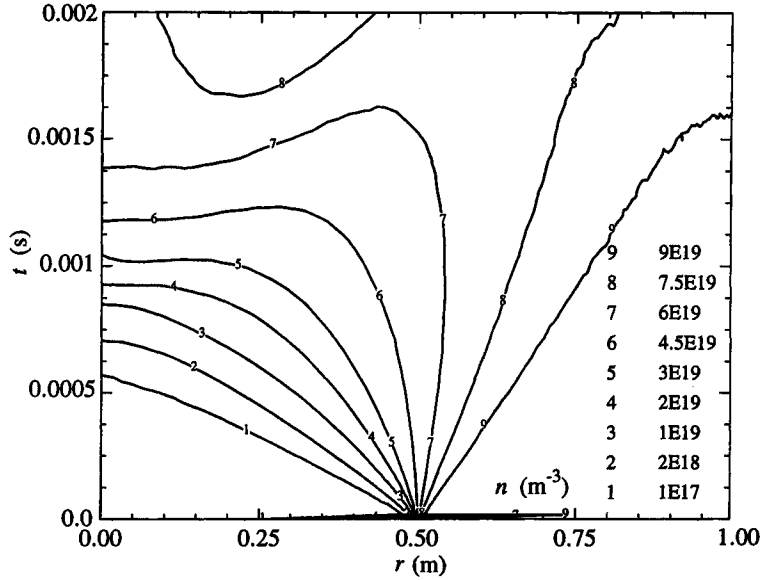


Fig. 13.19 Number densities in the collapse of a cylindrical cavity.

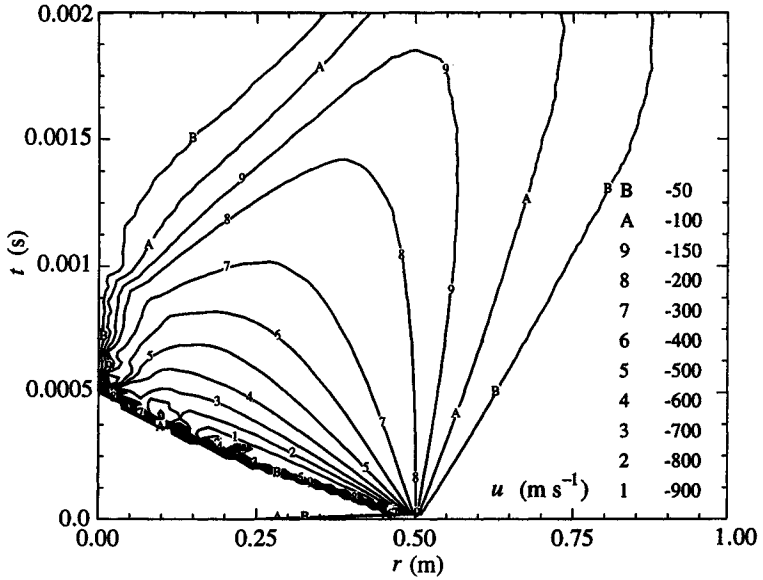


Fig. 13.20 Velocities in the collapse of a cylindrical cavity.

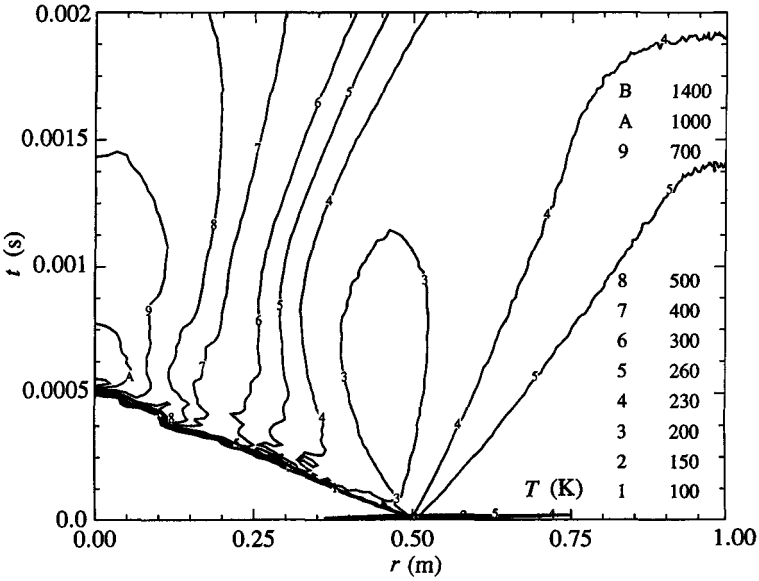


Fig. 13.21 Temperatures in the collapse of a cylindrical cavity.

The number density, velocity, and temperature contours are shown in Figs 13.19 to 13.21, respectively. The effective 'edge' of the gas moves at a speed of about 1000 ms^{-1}, and this is higher than the theoretical limiting velocity of three times the undisturbed speed of sound. This undisturbed speed of sound is 308 ms^{-1} and there is a region in which the flow speed exceeds 900 ms^{-1}. There is an almost stationary region in which the temperature is less than 200 K, but low temperatures are otherwise confined to the 'edge region' which is at a number density less than 1% of the undisturbed number density. The reflected waves that result from the interaction of the rarefaction with the decreasing area are compression waves and, as long as they remain the hypersonic flow near the edge of the expanding gas, these waves also move towards the axis. This causes the 'edge' to acquire some of the characteristics of a shock wave, and the temperature within a radius of one half the initial radius of the cavity is larger than the initial temperature. The maximum temperature at the axis is about 1500 K and occurs shortly after the initial arrival of the gas.

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WHITHAM, G.B. (1958). On the propagation of shock waves through regions of non-uniform area or flow. *J. Fluid Mech.* **4**, 337–360.